

SAP AUTOMATION INTELLIGENCE

Enterprise Product Blueprint & Complete Solution Design

Discover what is new. Understand how it works. Identify what to automate. Decide what to build.

Design basis: the supplied Vibe Coding guide recommends six pre-code documents—PRD, TRD, App Flow, UI/UX Design Brief, Backend Schema and Implementation Plan—to remove ambiguity before an AI coding agent starts work.

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Scope: this is a target-state enterprise design, not an MVP. It expands that six-document discipline into the complete 10-capability platform requested.

1. Executive Vision

Product name: SAP Automation Intelligence Engine (SAIE)

SAIE continuously monitors the SAP ecosystem, detects meaningful changes, extracts automation patterns, reconstructs technical architecture, validates evidence, identifies business opportunities, scores buildability and produces recurring intelligence reports.

The central question is not “What did SAP publish?” It is: “What changed, what automation pattern does it reveal, where can it be applied, and what should we build or replace because of it?”

Primary users

- SAP enterprise and solution architects.
- SAP automation / CoE leaders.
- SAP migration and clean-core teams.
- BTP, integration and development teams.
- Industry solution teams.
- Innovation, consulting and product leadership.

Primary outcome

A continuously updated, evidence-backed repository of SAP automation intelligence plus a ranked backlog of reusable automation and product opportunities.

2. Capability 01 — Discovery & Source Intelligence

- Maintain a Source Registry with source ID, URL, type, domain, industry, priority, schedule, last crawl, content hash and active status.
- Use source-specific acquisition: HTML, RSS/API, structured documentation and sitemap discovery where available.
- Run continuous ingestion while using Saturday as the six-day reporting boundary.
- Tier sources by authority: SAP documentation, SAP announcements, SAP Community, partner/customer evidence and general discovery.
- Preserve URL, retrieval timestamp and content version for every finding.
- Detect source-level changes before expensive semantic analysis.
- Respect robots.txt, terms, rate limits and authentication boundaries.

3. Capability 02 — Change Detection & Evidence Intelligence

- Store policy-compliant source versions or normalized snapshots.
- Generate lexical and semantic diffs.
- Classify changes as new capability, enhancement, documentation clarification, deprecation, architecture change, event announcement or no meaningful change.
- Assign evidence confidence based on authority, recency, corroboration and specificity.
- Merge multiple sources into one canonical finding.
- Label facts as confirmed, inferred or speculative.
- Maintain an evidence trail for every priority finding.

4. Capability 03 — Automation Pattern Intelligence

- Extract business process, business area, industry, SAP products, capability, trigger, inputs, decisions, workflow, human involvement and outcome.
- Classify workflow, RPA, document processing, API integration, event-driven, AI-assisted, agentic, predictive or custom automation.
- Capture the business problem and pre-automation process.
- Record benefits only when stated; mark inferred benefits explicitly.
- Map findings across FI/CO, SD, MM, PP, QM, PM/EAM, EWM, TM, PS, PLM, MDG, GRC, Treasury, CRM, HCM, procurement and industries.
- Assign canonical Automation IDs and temporal lineage.

5. Capability 04 — Architecture Reconstruction

- Extract trigger → data source → processing → AI/rules → decision → workflow → API/event → target → monitoring.
- Identify SAP technologies when evidenced: S/4HANA, BTP, Integration Suite, APIs, Event Mesh, Build, Process Automation, AI, Data Cloud, Datasphere, HANA, Cloud ALM and related services.
- Separate confirmed from inferred components.
- Generate logical architecture diagrams and text summaries.
- Identify synchronous API, asynchronous event, batch, workflow, document and agent invocation patterns.
- Capture human-in-loop controls, approvals and exception paths.
- Flag security, audit, resilience, observability and data-governance considerations for validation.

6. Capability 05 — Opportunity Validation

- Check whether the discovery is standard, configurable, extensible, partner-provided or genuinely missing.
- Map capability to customer process pain and manual effort.
- Identify ECC-to-S/4 and clean-core implications.
- Classify path: standard SAP, configuration, extension, BTP automation, custom app, AI agent or external integration.
- Assess reuse across customers and industries.

- Track dependencies on release, edition, licensing and services.
- Create a human validation checklist.

7. Capability 06 — Scoring & Prioritization

- Score Business Value, Automation Potential, Technical Feasibility, Reusability, Demand, Differentiation and Clean-Core relevance.
- Apply a transparent complexity penalty.
- Store numerical score plus rationale.
- Allow reviewer override with an explanation.
- Rank opportunities for the Saturday report and backlog.

8. Capability 07 — Knowledge Graph & Intelligence Repository

- Link sources, findings, automation patterns, products, processes, industries, technologies, APIs, events, architectures and opportunities.
- Support temporal queries such as what changed in the last 30/90/180 days.
- Support cross-domain queries such as AI automation affecting MM and manufacturing.
- Store evidence and confidence at fact/relationship level where practical.
- Maintain lineage from source → extraction → validation → score → report.
- Provide semantic search plus structured filters.
- Enable related-pattern and reusable-architecture recommendations.

9. Capability 08 — Reporting & Saturday Intelligence

- Generate executive summary, meaningful changes, automation findings and top opportunities.
- Provide detailed Automation Cards with architecture and evidence.
- Explain what changed and why it matters.
- Include domain, industry and technology heat maps.
- Include ECC-to-S/4 and clean-core opportunity flags.
- Generate PDF, HTML and JSON/CSV exports.
- Allow configurable recipients, filters, scoring weights and schedule.

10. Capability 09 — User Experience, Governance & Operations

- Provide Dashboard, Discovery, Automation, Architecture, Opportunity, Evidence, Reports and Administration workspaces.
- Provide RBAC, audit logs, model/prompt versioning and evidence traceability.
- Monitor source health, agent health, retries, dead-letter queues and cost budgets.
- Route low-confidence and high-impact items to human review.
- Enforce tenant isolation and least privilege.
- Provide alerts and operational runbooks.

11. Capability 10 — Continuous Improvement & Productization

- Capture reviewer feedback on relevance, correctness, architecture and scoring.
- Maintain quality metrics and benchmark datasets.
- Continuously evolve taxonomies and scoring calibration.
- Support customer-specific source packs and private knowledge spaces.
- Expose future APIs for automation-opportunity and architecture retrieval.
- Future recommendation layer: what to build, what to replace and what reusable accelerator to create.

12. Document 01 — PRD

The supplied guide calls the PRD the north star and recommends the problem, target user, value proposition, features, user stories, out-of-scope items and success metrics. □filecite□turnOffile0□L30-L59□

Product objective

Create a trusted SAP automation intelligence system that continuously discovers ecosystem changes and converts them into actionable, evidence-backed automation and product opportunities.

Core features

- Source registry and scheduled acquisition.
- Change detection and semantic deduplication.
- Automation extraction and classification.
- Architecture reconstruction.
- Evidence/provenance management.
- Opportunity validation and scoring.
- Knowledge graph and semantic search.
- Interactive intelligence workspace.
- Saturday report generation.
- Continuous learning and governance.

Representative user stories

- As an SAP architect, I want new automation capabilities by business domain so I can evaluate relevance.
- As an automation lead, I want ranked opportunities so I can prioritize reusable solutions.
- As a migration architect, I want clean-core and ECC-to-S/4 implications so I can identify replacement opportunities.
- As a technical architect, I want evidence-backed architecture extraction so I can understand how a capability works.
- As a practice leader, I want a Saturday report so I can brief stakeholders without manually scanning dozens of sources.
- As a reviewer, I want to correct classifications and scores so the system improves.

Out of scope / controlled boundaries

- Autonomous production deployment of customer SAP changes.
- Unreviewed claims presented as confirmed SAP facts.
- Bypassing authentication, robots.txt, paywalls or site controls.
- Production code generation without architecture/security review.
- Treating generic web commentary as authoritative technical evidence.

Success metrics

- ≥85% precision for genuinely-new findings after human evaluation.
- ≥80% relevance for automation-related findings.
- ≥90% duplicate consolidation for multi-source coverage.
- ≥90% of priority findings have traceable evidence.
- ≥80% of reviewed architecture summaries judged useful by SAP architects.
- Improving human acceptance rate of top-ranked opportunities.
- Saturday report generated on schedule and auditable.

13. Document 02 — TRD

The reference guide defines the TRD as the technical blueprint for framework, backend, database, authentication, hosting, APIs, libraries, environment variables and constraints. `filecite``turn0file0``L62-L93`

Layer	Target technology	Purpose
Web UI	Next.js + TypeScript	Enterprise workspace
API	Python FastAPI	Typed service APIs
Agent orchestration	Workflow/orchestration framework	Controlled multi-agent execution
Workers	Python async workers	Crawl/enrich/report jobs
Database	PostgreSQL	Transactional metadata
Vector	pgvector / vector store	Semantic retrieval
Graph	Neo4j or graph model	Relationships
Object storage	S3-compatible	Snapshots/reports
Queue	Kafka / managed queue / Redis Streams	Decoupled processing
Search	OpenSearch / PostgreSQL FTS	Search and facets
LLM	Enterprise model gateway	Extraction/reasoning
Observability	OpenTelemetry	Metrics/logs/traces
Deployment	Kubernetes / managed containers	Scalable services
Identity	OIDC/SAML-capable IdP	SSO/RBAC

Technical principles

- Evidence-first and provenance-preserving.
- Deterministic preprocessing before generative reasoning.
- Human-in-the-loop for high-impact decisions.
- Model-agnostic LLM gateway.
- Asynchronous event-driven processing.
- Idempotent, replayable jobs.
- Version prompts, models and classifiers.
- Least privilege and tenant isolation.
- Secure secret management.
- Full auditability.

14. Document 03 — App Flow

The source guide says app flow should map screens, navigation, entry points, authentication, core journeys, edge cases and redirects. `□filecite□turn0file0□L96-L125□`

Area	Pages
Home	Overview / Weekly Pulse
Discovery	Sources / Crawl Runs / Changes / Errors
Intelligence	Automation Findings / Detail / Architecture
Opportunities	Backlog / Detail / Validation
Knowledge	Products / Processes / Industries / Technologies / APIs
Reports	Saturday Reports / Detail / Export
Governance	Review Queue / Evidence / Audit
Administration	Users / Sources / Schedules / Scoring / Agents / Models
Settings	Profile / Notifications / Preferences

Core journeys

Discover: Dashboard → New Changes → Finding → Evidence → Automation Card → Architecture → Opportunity → Backlog.

Evaluate: Opportunity Backlog → Filter → Detail → Evidence → Architecture → Score → Assign → Validate/Reject.

Report: Reports → Current six-day report → Executive summary → Top opportunities → Detailed findings → Evidence → Export.

Agent flow

Scheduler → Acquisition → Parse → Normalize → Hash/Diff → Relevance → Deduplicate → Automation Extraction → Architecture → Evidence → Opportunity → Scoring → Knowledge Graph → Report → Notification.

Edge cases

- Unavailable source: retry/backoff, then alert.
- Changed page structure: quarantine parser result.
- Conflicting sources: retain claims and reduce confidence.
- Duplicate: merge under canonical finding.
- LLM failure: retry/fallback/review.
- Low confidence: do not promote automatically.
- Report failure: retry and alert; never publish incomplete report.

15. Document 04 — UI/UX Design Brief

The supplied guide recommends defining visual direction, palette, typography, component style, responsiveness, accessibility and reference patterns. □filecite□turn0file0□L128-L161□

Design direction

Modern enterprise intelligence console: analytical, trustworthy, dense but readable, evidence-first, with clear separation between confirmed facts and AI inference.

Core patterns

- Collapsible left navigation.
- Executive KPI cards.
- Dense sortable/filterable findings tables.
- Evidence badges: Confirmed / Corroborated / Inferred / Speculative.
- Transparent score badges.
- Clickable architecture diagrams.
- Source-change timeline.
- Evidence side drawers.
- Human review queue.
- Responsive laptop/tablet layout.

Accessibility

- Keyboard navigation and strong contrast.
- Do not encode meaning by color alone.
- Accessible tables and focus states.
- Readable typography and configurable density.

Primary screens

- Weekly Pulse Dashboard
- Change Explorer
- Automation Intelligence Board
- Automation Detail
- Architecture Viewer
- Opportunity Radar
- Evidence Explorer
- Report Viewer
- Review Queue
- Source Health
- Agent Operations
- Administration

16. Document 05 — Backend Schema

The source guide emphasizes tables, columns, keys, relationships, indexes, auth, roles, sensitive fields, storage, events and APIs. [filecite](#)[turn0file0](#)[L164-L197](#)

Entity	Important fields	Relationships
tenants	id, name, settings	users, sources, findings
users	id, tenant_id, email, role	reviews, audit
sources	id, url, type, priority, schedule	crawl_runs, versions
crawl_runs	id, source_id, status, metrics	source, versions
source_versions	id, source_id, hash, retrieved_at	source, changes
changes	id, version_id, type, confidence	findings
findings	id, canonical_key, title, status	changes, automations
automations	id, finding_id, domain, industry, type	architecture, opportunities
architecture_nodes	id, automation_id, node_type, name	edges
architecture_edges	id, automation_id, from, to, relation	nodes
evidence	id, finding_id, source_id, locator, confidence	finding
opportunities	id, automation_id, status, score, owner	scores
scores	id, opportunity_id, metric, value, rationale	opportunity
reports	id, tenant_id, period, status, file_uri	report_items
report_items	id, report_id, finding_id, rank	reports/findings
reviews	id, entity, reviewer, decision, comments	users
agent_runs	id, agent_type, model, prompt, status	audit
audit_log	id, actor, action, entity, timestamp	users

Security model

- Tenant isolation at every query boundary.
- RBAC: platform_admin, tenant_admin, architect, analyst, reviewer, executive, read_only.
- Audit administrative and review actions.
- Encrypt secrets and sensitive configuration outside application data.
- Apply source-content retention and licensing policies.

17. Document 06 — Enterprise Implementation Plan

The supplied guide recommends explicit phases, dependencies and done criteria; it specifically calls for setup, database, authentication, core features, UI polish, testing and deployment. [filecite0turn0file0L200-L239](#)

Phase	Goal	Done criteria
1	Product & Architecture	Approved PRD, taxonomy, security and target-state architecture
2	Platform Foundation	Repo, CI/CD, environments, identity and observability
3	Data Foundation	DB, vector, graph, storage and migrations
4	Source Intelligence	Reliable replayable ingestion and source health
5	Change Intelligence	Versioning, diff, classification and deduplication
6	Automation Intelligence	Structured Automation Cards and taxonomy mapping
7	Architecture Intelligence	Evidence-backed architecture graph and diagrams
8	Opportunity Intelligence	Validated, scored opportunity backlog
9	Knowledge & Search	Entity resolution, graph, semantic/faceted search
10	Workspace UX	End-to-end dashboard and workspaces
11	Reports	Six-day aggregation, Saturday PDF/HTML/JSON and alerts
12	Governance	Review, audit, prompt/model registry
13	Evaluation	Golden sets, regression, precision/recall and quality gates
14	Production Hardening	HA, DR, security, cost controls and runbooks
15	Continuous Learning	Feedback, calibration and source optimization
16	Productization	Tenant packs, APIs and external product foundation

18. Multi-Agent Operating Model

Agent	Responsibility	Output
Discovery	Find new/changed ecosystem content	Candidate items
Evidence	Validate authority/corroboration	Evidence package
Change	Determine material change	Change record
Automation	Extract business problem/pattern	Automation Card
Architecture	Reconstruct technical design	Architecture graph
Opportunity	Determine applicability/build path	Opportunity assessment
Scoring	Rank consistently	Score + rationale
Knowledge	Resolve entities/update graph	Relationships
Report	Create Saturday package	PDF/HTML/JSON
Review	Route uncertain/high-impact items	Review queue
Governance	Enforce source/model/policy controls	Audit/policy events

Operating rule: agents communicate through typed contracts and persistent artifacts. No agent should silently invent facts, directly deploy production changes, or bypass governance.

19. Opportunity Scoring Framework

Metric	Weight	Meaning
Business value	20%	Operational/financial/customer impact
Automation potential	15%	Manual work removable
Technical feasibility	15%	APIs, events, data, supported path
Reusability	15%	Cross-customer/industry potential
Demand	10%	Evidence of recurring need
Differentiation	10%	Distinctive value
Clean-core relevance	10%	Modernization/replacement potential
Complexity penalty	Up to -15%	Delivery and operational difficulty

Scores are recommendations, not facts. Store both the score vector and rationale; allow reviewer override with reason.

Opportunity classes

- Adopt standard SAP capability.
- Configure existing capability.
- Extend SAP clean-core.
- Build BTP automation.
- Build reusable integration accelerator.
- Build AI/agent solution.
- Build industry-specific product.
- Monitor only — insufficient evidence or low value.

20. Saturday Report Specification

Report title: SAP Automation Intelligence — Week ## / YYYY

Executive section

- Sources scanned and source health.
- Meaningful changes.
- Unique automation patterns.
- Top opportunities.
- Industry and technology signals.
- Clean-core / ECC-to-S/4 opportunities.
- Items requiring architect review.

Detailed Automation Card

Finding ID • Title • First detected • Last updated • Domain • Industry • SAP products • Automation type • Problem • Current process • New capability • Approach • Technology • Architecture • Human involvement • Benefits • Evidence • Confidence • Opportunity score • Recommended action.

Appendices

- All findings.
- Rejected/duplicate findings.
- Source changes.
- Evidence index.
- Scoring methodology.
- Agent/model versions.
- System health summary.

21. Source Strategy & SAP Coverage

The supplied request covers SAP master catalogs, S/4HANA functional domains, procurement, supply chain, manufacturing, CRM/CX, retail/media, BTP/automation/development, industries, technical innovation and migration/clean-core resources. The platform should preserve that breadth.

Monitoring hierarchy

- Tier 1: SAP Help / What's New / official product documentation.
- Tier 2: SAP News and official SAP innovation/product pages.
- Tier 3: SAP Community and official technical blogs.
- Tier 4: SAP Business Accelerator Hub, APIs and events.
- Tier 5: SAP Learning and architecture content.
- Tier 6: SAP industry and innovation pages.
- Future: partners, customer cases, GitHub, conferences and other ecosystem sources.

Functional taxonomy

FI/CO, SD, MM, PP, QM, PM/EAM, EWM, TM, PS, PLM, MDG, GRC, Treasury, Real Estate, Service, CRM/CX, Ariba, Concur, HCM/SuccessFactors, supply chain, manufacturing, retail, media, life sciences, healthcare and the wider SAP industry portfolio.

Technology taxonomy

S/4HANA, BTP, Integration Suite, API Management, Event Mesh, Process Automation, Build, Work Zone, ABAP, CAP, Business Application Studio, Data Cloud, Datasphere, HANA, AI services, Joule, Cloud ALM, APIs, events and migration technologies.

22. Risks, Controls & Non-Functional Requirements

Risk	Control
Hallucinated architecture	Evidence-linked extraction; confirmed/inferred labels; review
Duplicate findings	Canonical topic/entity resolution
Stale information	Versioning and change timestamps
Crawler breakage	Source health monitoring and parser quarantine
LLM cost growth	Tiered models, caching, deterministic preprocessing, budgets
Model lock-in	Model gateway and versioned abstraction
Security	SSO, RBAC, tenant isolation, secrets, audit
Source/legal risk	Respect robots, terms, licenses and retention
False opportunity ranking	Transparent scoring + reviewer override
Operational failure	Queues, retries, DLQ, observability and runbooks

Non-functional targets

- Auditable: every report finding traceable to evidence.
- Reliable: scheduled Saturday report with retries and alerts.
- Scalable: ingestion and analysis workers independently scalable.
- Secure: SSO, least privilege and tenant isolation.
- Observable: crawl, agent, cost, latency and quality metrics.
- Maintainable: modular services, typed contracts and versioned agents.
- Recoverable: replayable ingestion and report jobs.
- Explainable: score and confidence rationale available.

23. Final Product Definition

SAP Automation Intelligence Engine is an enterprise multi-agent intelligence platform that continuously observes the SAP ecosystem and converts ecosystem change into structured automation knowledge and prioritized build opportunities.

Target-state chain: Discover → Verify → Detect Change → Understand Automation → Reconstruct Architecture → Validate Opportunity → Score → Connect Knowledge → Report → Learn → Recommend What to Build.

The supplied reference says that once the six documents are complete, they should be given to the AI coding agent as the source of truth for everything it builds. [filecite\turn0file0\L241-L255](#) This master design is intended to serve that role, with the additional enterprise architecture required by the 10-section solution.

Next engineering package

Before implementation begins, produce a Build Specification Pack containing exact database DDL, API contracts, event schemas, agent contracts/prompts, source connector specifications, scoring formulas, UI screen specifications, report templates, security controls and deployment topology.